

Particles and Nuclei 2

- 2/10/26
- Announcements:
- HW 2 Due this Thurs in class 2/12/26
- Still No further reading for Thurs - slides/ HW

List of Kinematics Topics

II. PLEASE REMEMBER

(masses are usually 0!)

A. $p \equiv (E, \vec{p})$ is a 4-vector

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$\left. \begin{array}{l} D. \quad p_{||} = m_T \sinh y \\ E = m_T \cosh y \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} m_T = \sqrt{m^2 + p_T^2} \\ y = \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{array} \right.$$

Rapidity

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

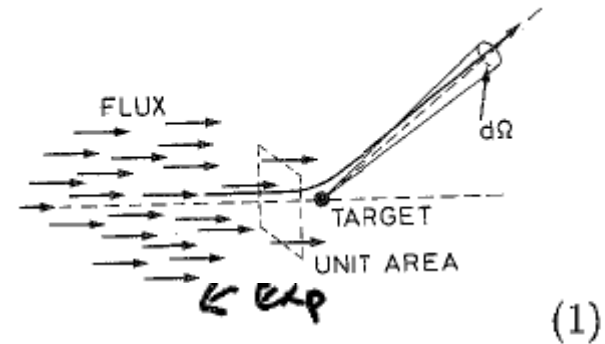
this is
9
Cross
Section
↓
Invariant
Cross Section

Cross Section x.s.

- $d\sigma/d\Omega$ might be the more fundamental quantity than σ
 - **Think of this as x.s. as a fn of $\phi, \theta \rightarrow xs(\phi, \theta)$** “scattering amount in given direction”
 - usually just as a function of θ
- Total Cross section: $\sigma = \int d\sigma \rightarrow$ “fraction” (effective beam direction cross sectional area) of incident luminosity that experiences scattering $\rightarrow N_{scatt}^{total}$
- You are measuring the fraction that goes in the direction of various values of theta : ($\theta_i \rightarrow f_i$)

$$\frac{d\sigma}{d\Omega}(E, \Omega) = \frac{1}{F} \frac{dN_s}{d\Omega},$$

$$N_s = \frac{d\sigma}{d\Omega} N_0 n_T \Delta\Omega.$$



In general, $\frac{d\sigma}{d\Omega}$ depends on the type of projectile, the type of target, the energy of the projectile, and the scattering angle. Note also that the above formula is technically only exactly true in the limit that $\Delta\Omega \rightarrow 0$, due to the fact that $\frac{d\sigma}{d\Omega}$ depends on the scattering angle. The number of scattered particles N_s is directly proportional to N_0 , n_1 , and $\Delta\Omega$ as one intuitively expects – in this picture, the differential cross section is the proportionality constant. It is often said that the differential cross section “contains the physics”...

“Solid Angle” $d/d\Omega \rightarrow$ Angular Direction $\vec{\Omega}$

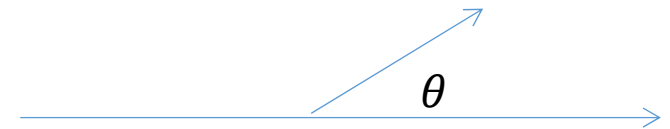
- A few “tricky” but actually simple quantities
- Solid angle : Ω serves double duty
 - 2 just short hand for **angular direction** $[\phi, \theta]$ (like a 2-D vector) $\vec{\Omega}$
 - “Solid angle” itself is $\Delta\Omega$ or $d\Omega = d\phi d(\cos\theta) = \sin\theta d\theta d\phi$ (angular area $\perp \vec{\Omega}$)
 - BUT NOTICE $d\sigma/d\Omega$ is NOT WRITTEN $d\sigma/d\Delta\Omega$
 - $d\sigma/d\Omega \propto dN_{\text{scatt}}/d\Omega = \frac{dN}{d\theta d\phi}$

confusion: it's like calling $dp_x dp_y$ as dA but $d\vec{A} \neq dx dy$!

Think of this as $x.s./n_{\text{scatt}}$ as a fn of $\phi, \theta \rightarrow xs(\phi, \theta)$

- Differential yields: $dN/dx \rightarrow$ (think of as) Particle yield (number) as a function of x But does obey all things like a derivative!

- $dN/dx = \int^{all y} dy dN/(dx dy)$
- $\int d\phi dN/d\Omega = dN/d\theta$
- $= \text{const } f(\theta) = 2\pi f(\theta)$
- Why most diff. x.s. only have θ



Cross Section $\rightarrow$ Just Counted N of something

- Cross Section (x.s.):

- Defined by equations like

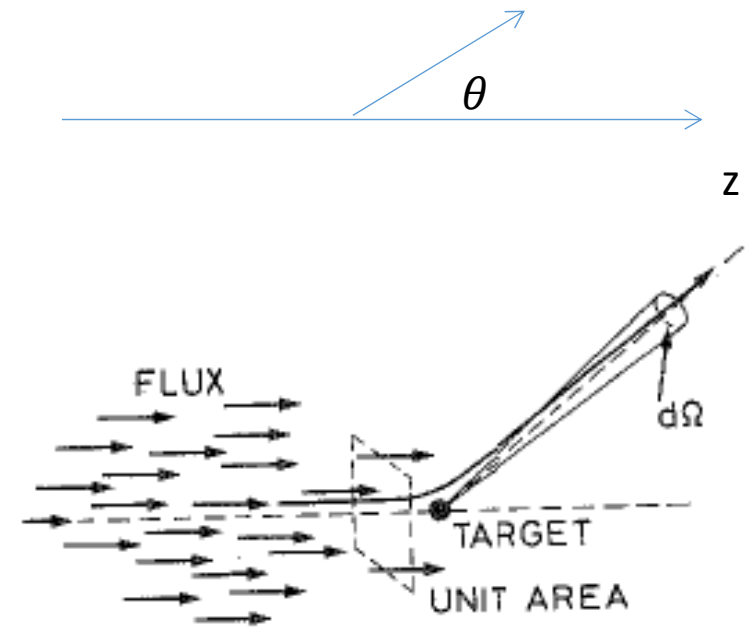
$$\text{Flux} * (\text{x.s.}) = \text{Scattering Yield}$$

- Luminosity / Flux F : particles per second per unit area in beam

- Often we integrate over time:
integrated luminosity / flux

- x.s. has units of area

Best to think of as weird normalization for scattered "yield"



$$\frac{d\sigma}{dx} \propto \frac{dn}{dx}$$

$$\frac{d\sigma}{d\Omega}(E, \Omega) = \frac{1}{F} \frac{dN_s}{d\Omega},$$

Normalization of Cross Section x.s.

We will discuss NORM FACTOR LATER →

∝ FLUX in single beam

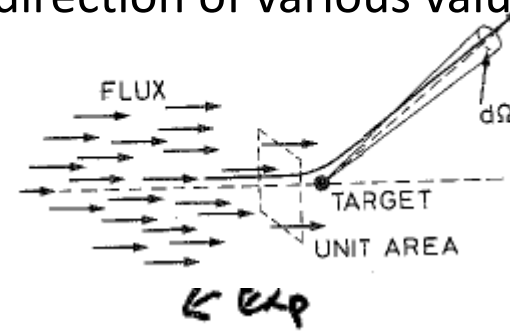
∝ LUMINOSITY collider

When we discuss high energy accelerators/colliders

- Total Cross section: $\sigma = \int d\sigma \rightarrow$ “fraction” (effective beam direction cross sectional area) of incident luminosity that experiences scattering $\rightarrow N_{scatt}^{total}$
- You are measuring the fraction that goes in the direction of various values of theta : $(\theta_i \rightarrow f_i)$

$$\frac{d\sigma}{d\Omega}(E, \Omega) = \frac{1}{F} \frac{dN_s}{d\Omega},$$

$$N_s = \frac{d\sigma}{d\Omega} N_0 n_T \Delta\Omega.$$



(1)

In general, $\frac{d\sigma}{d\Omega}$ depends on the type of projectile, the type of target, the energy of the projectile, and the scattering angle. Note also that the above formula is technically only exactly true in the limit that $\Delta\Omega \rightarrow 0$, due to the fact that $\frac{d\sigma}{d\Omega}$ depends on the scattering angle. The number of scattered particles N_s is directly proportional to N_0 , n_1 , and $\Delta\Omega$ as one intuitively expects – in this picture, the differential cross section is the proportionality constant. It is often said that the differential cross section “contains the physics”...

Differential Cross Section (or Distributions)

- So to summarize

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{d\Omega_p} = \frac{dn}{d\Omega_p} \propto \frac{dn}{d\theta_p d\phi_p} \left(\frac{1}{\sin\theta_p} \right)$$

- which is really just a sub part of :

$$\frac{dn}{d\vec{p}^3} = \frac{dn}{dp_x dp_y dp_z}$$

for d/dp^3
"directions" are meant

$$\begin{aligned} d\vec{p}^3 &= dp_x dp_y dp_z && \text{cartesian} \\ &= p^2 d\Omega_p = p^2 \sin\theta_p d\theta_p d\phi_p dp && \text{spherical} \\ &= p_{\perp} dp_{\perp} dp_z d\phi_p && \text{cylindrical} \end{aligned}$$

List of Kinematics Topics

II. PLEASE REMEMBER

(masses are usually 0!)

A. $p \equiv (E, \vec{p})$ is a 4-vector

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$D. \left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

show
cross
sections

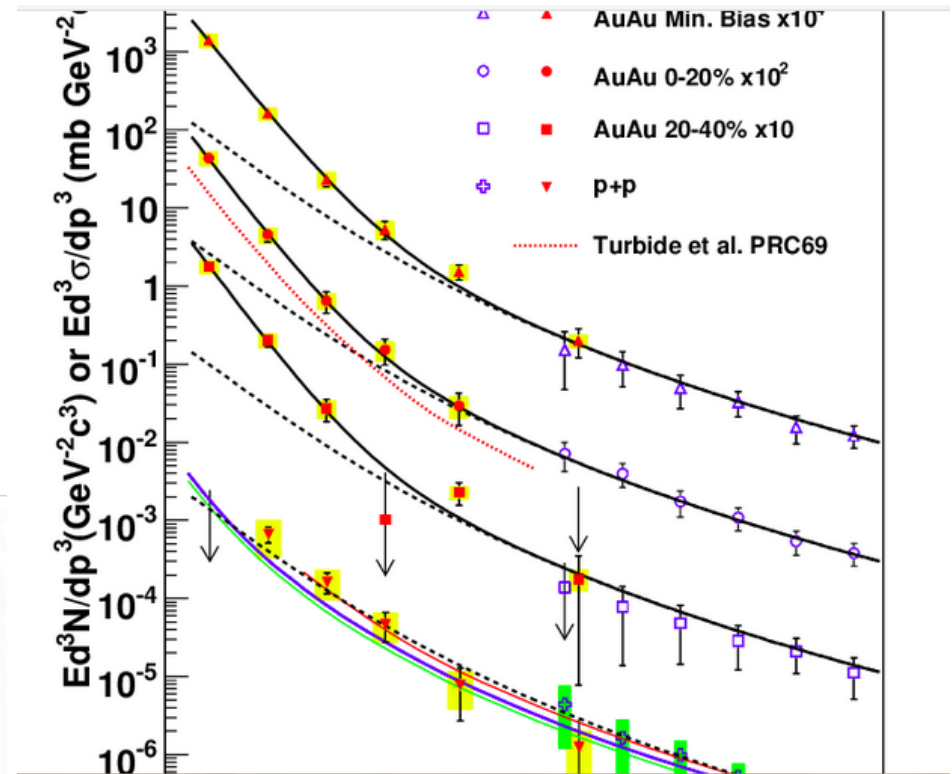
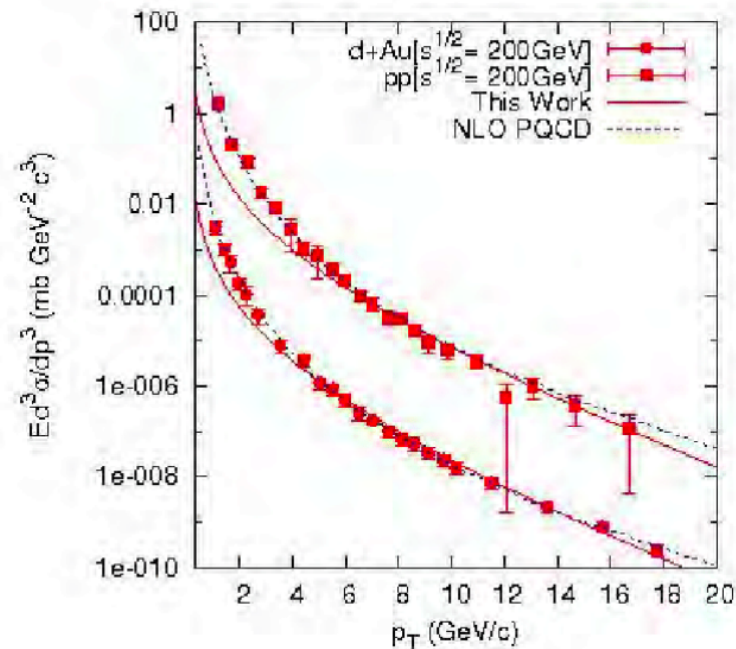
X.S. $\rightarrow$ Inv X.S.

$$\frac{dn}{d^3p/E} = \frac{dn}{dp_x dp_y \left(dp_z/E \right)}$$

$$d^3p = dp_x dp_y dp_z$$

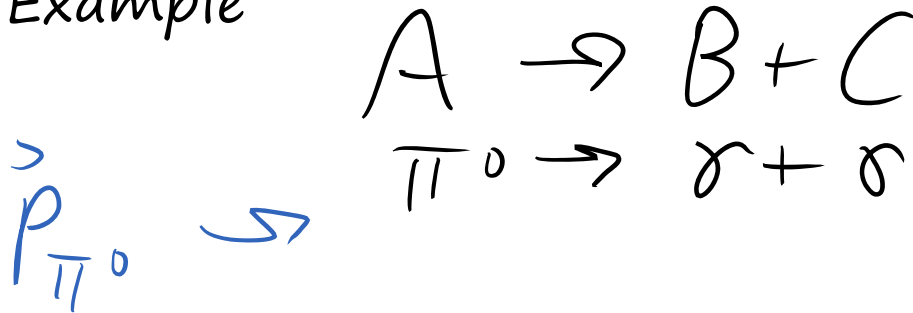
$$\propto \frac{dn}{dp_x dp_y \left(\frac{dp_z}{E} \right)} \rightarrow dy$$

Examples Published Inv Cross Sections.



Also can apply for DECAY distributions

- Example



$$\frac{dn}{d\theta_\gamma d\phi_\gamma} = ?$$

$$\frac{dn}{d\vec{p}^3/E} = \frac{dn}{dp_x dp_y (dp_z/E)}$$

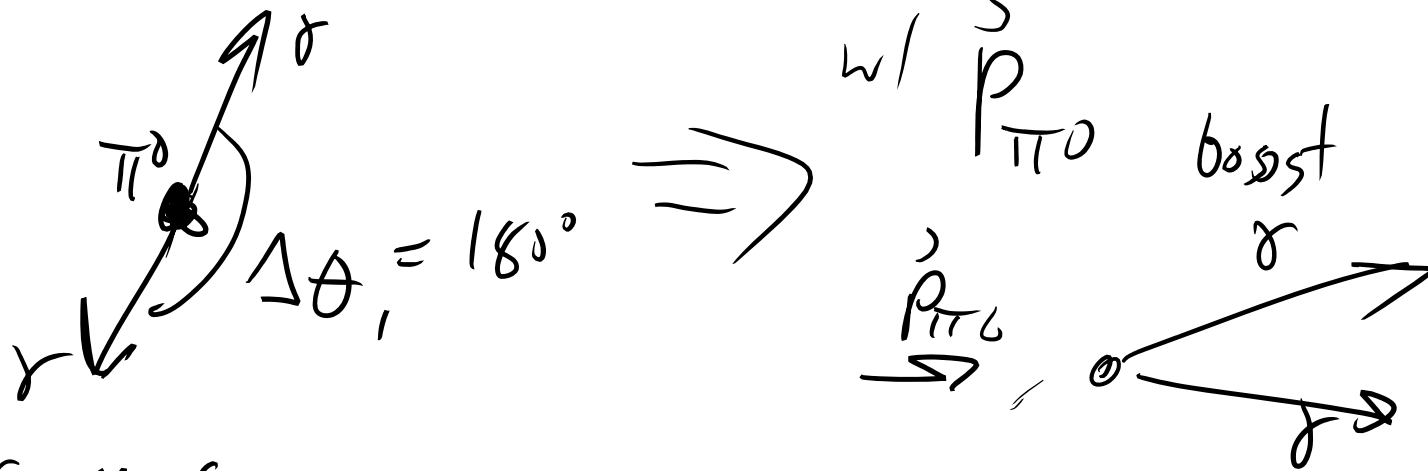
$$\Downarrow$$

$$\frac{dn_{\text{decay}}}{d\vec{p}_\gamma^3/E} = \text{same info}$$

How about boosted
decay distribution?
Given a relativistic $\vec{p}_{\pi^0}$

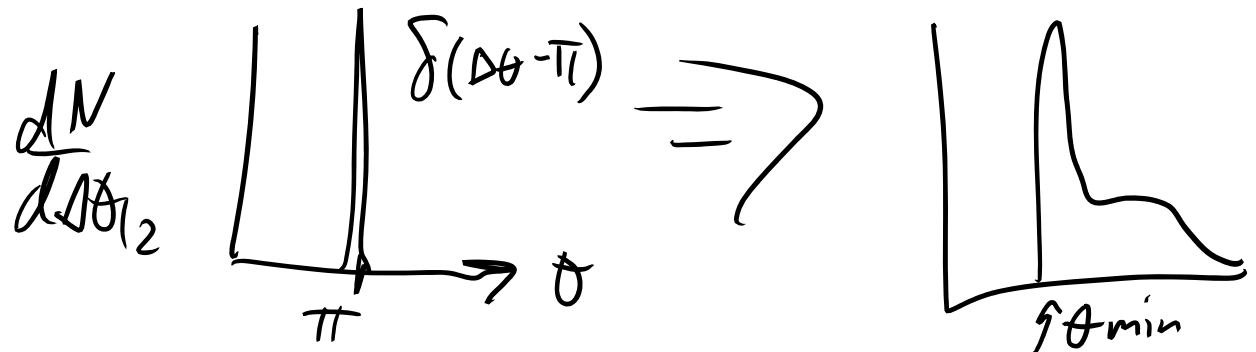
This ~~Next~~ time

- Boosting distributions.
- Based on how the hard sphere scattering is handled



COM frame

open angle $\Delta\theta_{12}$



Whole distribution?

- Exercise:

- Show $M = \sqrt{2E_1E_2} \sin(\theta/2)$

$$m^2 = p_1^\mu \cdot p_2^\mu = m^2 + m^2 - 2 p_1 \cdot p_2 \Rightarrow \sim 1 - \cos\theta$$

- And Show (by direct transformation of E_1^* E_2^*)

$$\sqrt{E_1E_2} = \frac{1}{2}\gamma M \sqrt{1 - \beta^2 \cos^2 \theta_1^*}$$

- (this means use these relations:)

$$E_1 = \frac{1}{2}\gamma M(1 + \beta \cos \theta_1^*)$$

$$E_2 = \frac{1}{2}\gamma M(1 - \beta \cos \theta_1^*).$$

Note that:

- 1) θ in the first expression really refers to the OPENING ANGLE btw the 2 photons (not polar angle —not difference of polar angles either—the theta as defined in the 3vect dot product)
- 2) θ_1^* in second expression **does** refer to polar angle of 1 in COM, more so specifying the “decay axis” direction

Whole Boosted Decay Opening Ang Distribution

- Note Griffiths Hard Sphere Scattering Example:
- To derive scat/decay distributions use derivatives!

$$\frac{dn}{d\Omega} \propto \frac{b}{\sin\theta} \left(\frac{db}{d\theta} \right) \quad b = R \cos\left(\frac{\theta}{2}\right) \quad \frac{db}{d\theta} = -\frac{1}{2} \sin\left(\frac{\theta}{2}\right)$$

Note that isotropic "emission" corresponds to $dn/d\cos\theta \propto 1$

(it's equal to some n-dependent constant actually, might as well choose 1)

- Using the same magic of differentials for us

$$\frac{dn}{d\Delta\theta} \equiv \frac{dn}{d\theta} \propto \frac{dn}{d\cos\theta^*} \left(\frac{d\cos\theta^*}{d\theta} \right)$$

from combining previous two exercise expressions

$$\cos\theta^* = \frac{1}{\beta} \sqrt{1 - \frac{1}{\gamma^2 \sin^2\theta/2}}$$

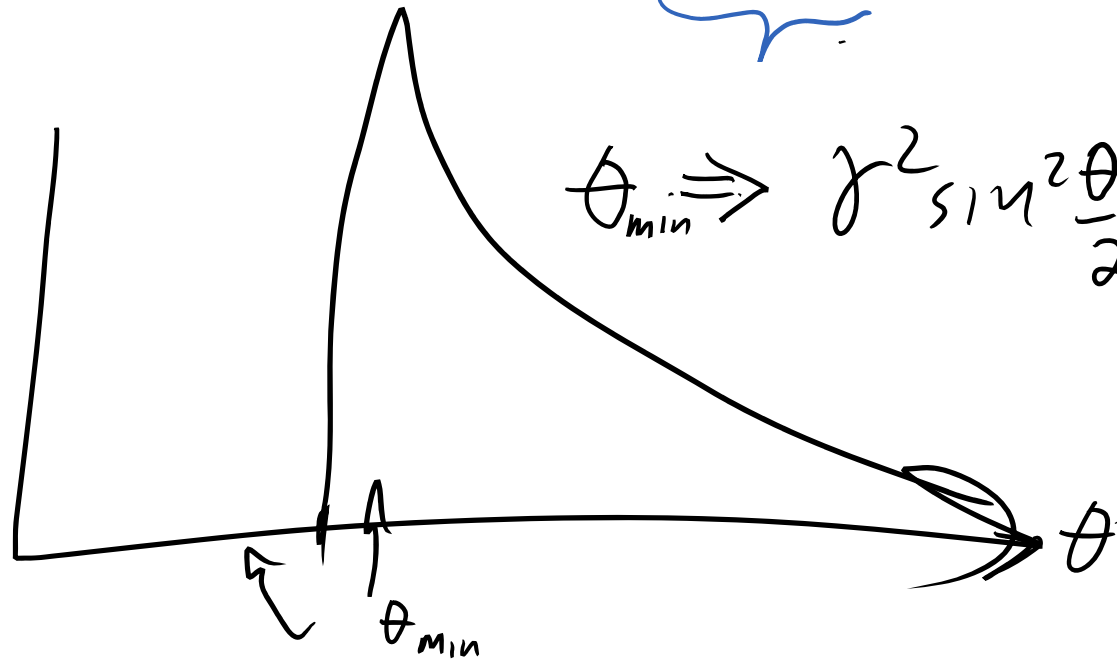
cont

- whole distribution (shape)

$$\frac{dn}{d\theta} \propto$$

$$\frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2} \left(\sqrt{\gamma^2 \sin^2 \frac{\theta}{2} - 1} \right)}$$

$$\frac{dn}{d\theta}$$



$$\theta_{min} \Rightarrow \gamma^2 \sin^2 \frac{\theta}{2} = 1$$

List of Kinematics Topics

II. PLEASE REMEMBER

(masses are usually 0!)

A. $p \equiv (E, \vec{p})$ is a 4-vector

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$D. \left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

show
cross
sections

Rapidity y and η (Pseudo-)

- Nucl lunch
- Two almost identical quantities that are useful partially because they both exist –
- use/derive properties of one applied to the other
- Note, it may seem from slides rapidity more related to Heavy Ion physics, but it is actually more of a COLLIDER thing
 - LHC [p+p] / HERA / Tevatron/ EIC communities all use this extensively!!!
 - Not sure why it's not covered in books, but I suspect they were ~all written by people who did much of their work before the prevalence of these highest energy colliders – and b.c. of the desire to connect to normal Quantum scattering coverage
 - Maybe also, only at these higher energy colliders did people start to explore more forward/backward angle physics –because it was inaccessible in fixed target
 - Also many initially important collider cross sections are strongly peaked at 90 degrees from beam where rapidity is not as important?

Start w/ Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

In experimental particle physics, **pseudorapidity**, a coordinate describing the angle of a particle relative to the beam axis as

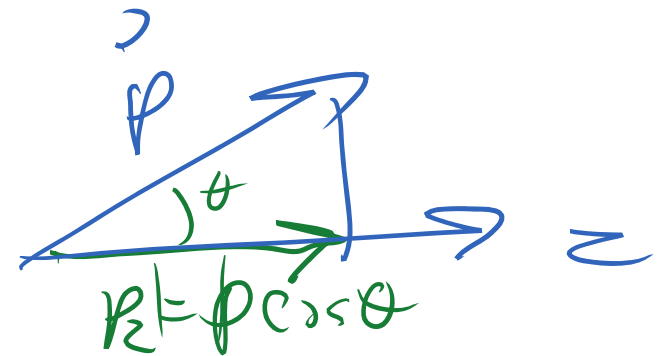
$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the angle between the particle three-momentum and the direction of the beam axis.^[1] Inversely,

$$\theta = 2 \arctan(e^{-\eta}).$$

As a function of three-momentum $\mathbf{p}$, pseudorapidity

$$\eta = \frac{1}{2} \ln \left(\frac{|\mathbf{p}| + p_L}{|\mathbf{p}| - p_L} \right) = \operatorname{artanh} \left(\frac{p_L}{|\mathbf{p}|} \right),$$



$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

- For massless particles (γ , $\sim \nu$, $\sim e \dots$) $y = \eta$ exactly
- Even for $\sim$ all massive particles they y, η almost identical except very VERY far forward and backward

Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

In experimental particle physics, **pseudorapidity**, a coordinate describing the angle of a particle relative to the beam axis as

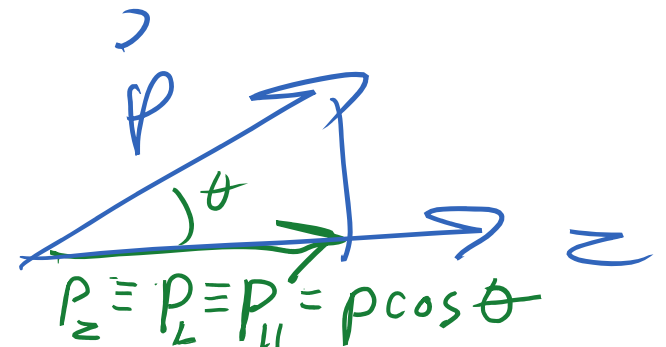
$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the angle between the particle three-momentum and the direction of the beam axis.^[1] Inversely,

$$\theta = 2 \arctan(e^{-\eta}).$$

As a function of three-momentum $\mathbf{p}$, pseudorapidity

$$\eta = \frac{1}{2} \ln \left(\frac{|\mathbf{p}| + p_L}{|\mathbf{p}| - p_L} \right) = \operatorname{artanh} \left(\frac{p_L}{|\mathbf{p}|} \right),$$



$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

Why same? $\Rightarrow$

$$= \frac{1}{2} \ln \left(\frac{\frac{1}{p}(p + p_L)}{\frac{1}{p}(p - p_L)} \right) \leftarrow \text{Same}$$

- For massless particles (γ , $\sim \nu$, $\sim e \dots$) $y = \eta$ exactly [see later]
- Even for $\sim$ all massive particles they y, η almost identical except very VERY far forward and backward

Equivalent Definitions of Pseudorapidity η

- z means z axis in coordinate system with z in beam direction. Theta, θ , is the polar angle in spherical coordinates (r, θ, ϕ) . $p = |\vec{p}|$ is the magnitude of the 3-mom

$$\eta = \frac{1}{2} \ln \left(\frac{p + p_z}{p - p_z} \right)$$

$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

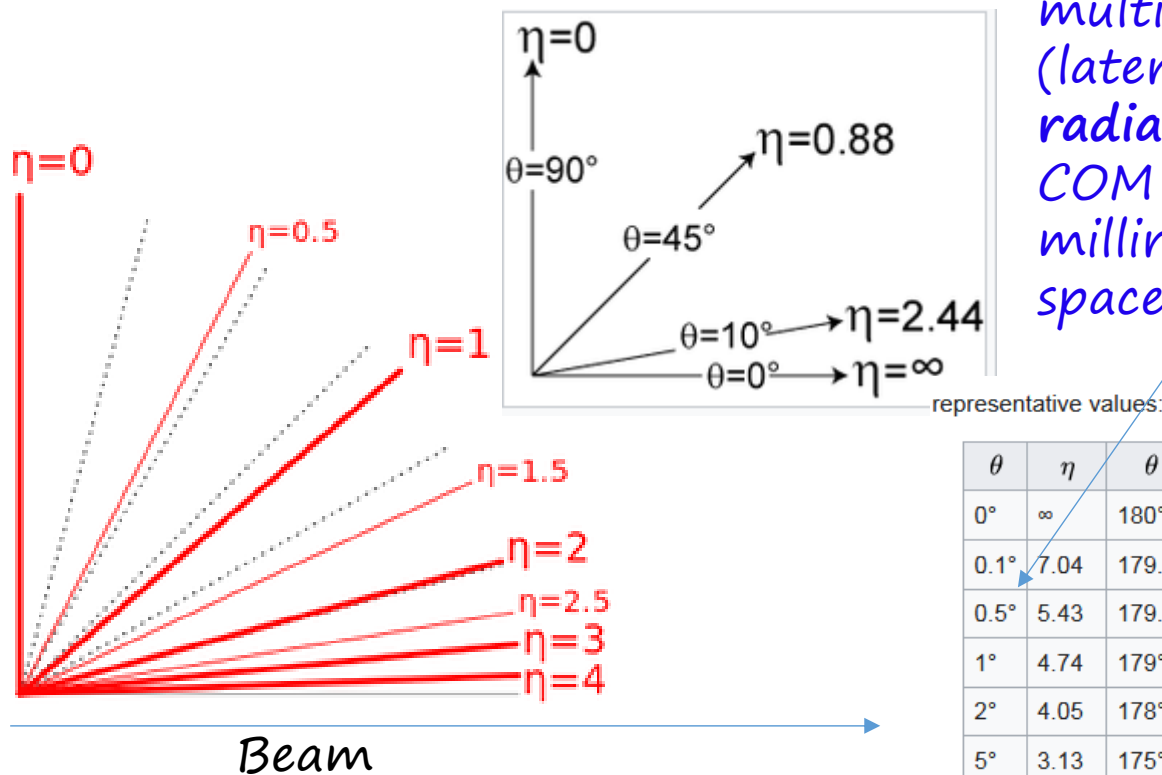
$$\eta = -\frac{1}{2} \ln \left(\tan \frac{\theta}{2} \right) \quad \eta = \frac{1}{2} \ln \left(\cot \frac{\theta}{2} \right)$$

$$\eta = \operatorname{arctanh}(p_z / p)$$

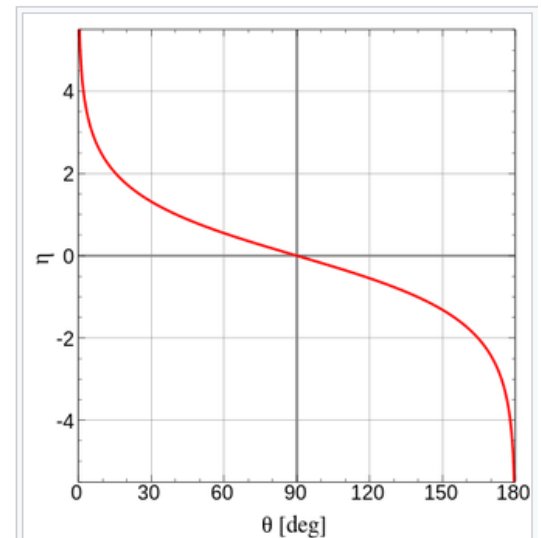
$$\eta \leftrightarrow \theta$$

- Pseudo rapidity allows an easier mapping to θ and shows us why this version of θ is so much better
- (From Wikipedia:pseudorapidity):

More than simply " $\ln\theta$ " : Close to beam, boost scrunches up multiple units of rapidity (later slides: = multiple radians [1 rad = 60 deg] of COM angular space!) in milliradians of lab angular space!!!



θ	η	θ	η
0°	∞	180°	$-\infty$
0.1°	7.04	179.9°	-7.04
0.5°	5.43	179.5°	-5.43
1°	4.74	179°	-4.74
2°	4.05	178°	-4.05
5°	3.13	175°	-3.13
10°	2.44	170°	-2.44
20°	1.74	160°	-1.74
150°	-1.32		



Particle Physics Rapidity y

- Note there is another slightly different definition of rapidity y (a different, but related, quantity) in textbooks. The one we are interested in is the “particle physics” / collider version
- Experimentally, for this more usual particle physics kind, we typically **ONLY** consider boosts (& ll direction as being) along the (common) **beam direction p_z -- common to all particles**
 - where as the other version is in each particle's direction
- Our version will be defined similarly to the pseudo-rap:

$$y = \frac{1}{2} \ln \left(\frac{E + p_{z/\text{beam}}}{E - p_{z/\text{beam}}} \right)$$


where z is the “global” coordinate z along the beam direction

- **Example :** what will y be for a particle moving perpendicular to beam (ie with $\theta = 90$ deg)
 - In that case $p_z = 0$
 - $y = \frac{1}{2} (\ln (E + 0/E - 0)) = \frac{1}{2} \ln 1 = 0$

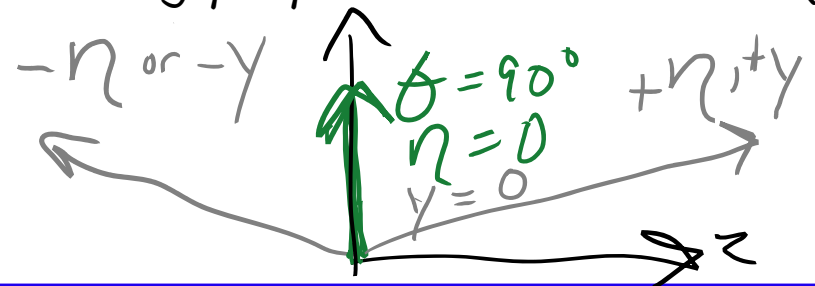
Particle Physics Rapidity y

- Note there is another slightly different definition of rapidity y (a different, but related, quantity) in textbooks. The one we are interested in is the “particle physics” / collider version
- Experimentally, for this more usual particle physics kind, we typically **ONLY** consider boosts (& || direction as being) along the (common) **beam direction p_z -- common to all particles**
 - where as the other version is in each particle's direction
- Our version will be defined similarly to the pseudo-rap:

$$y = \frac{1}{2} \ln \left(\frac{E + p_{z/\text{beam}}}{E - p_{z/\text{beam}}} \right)$$

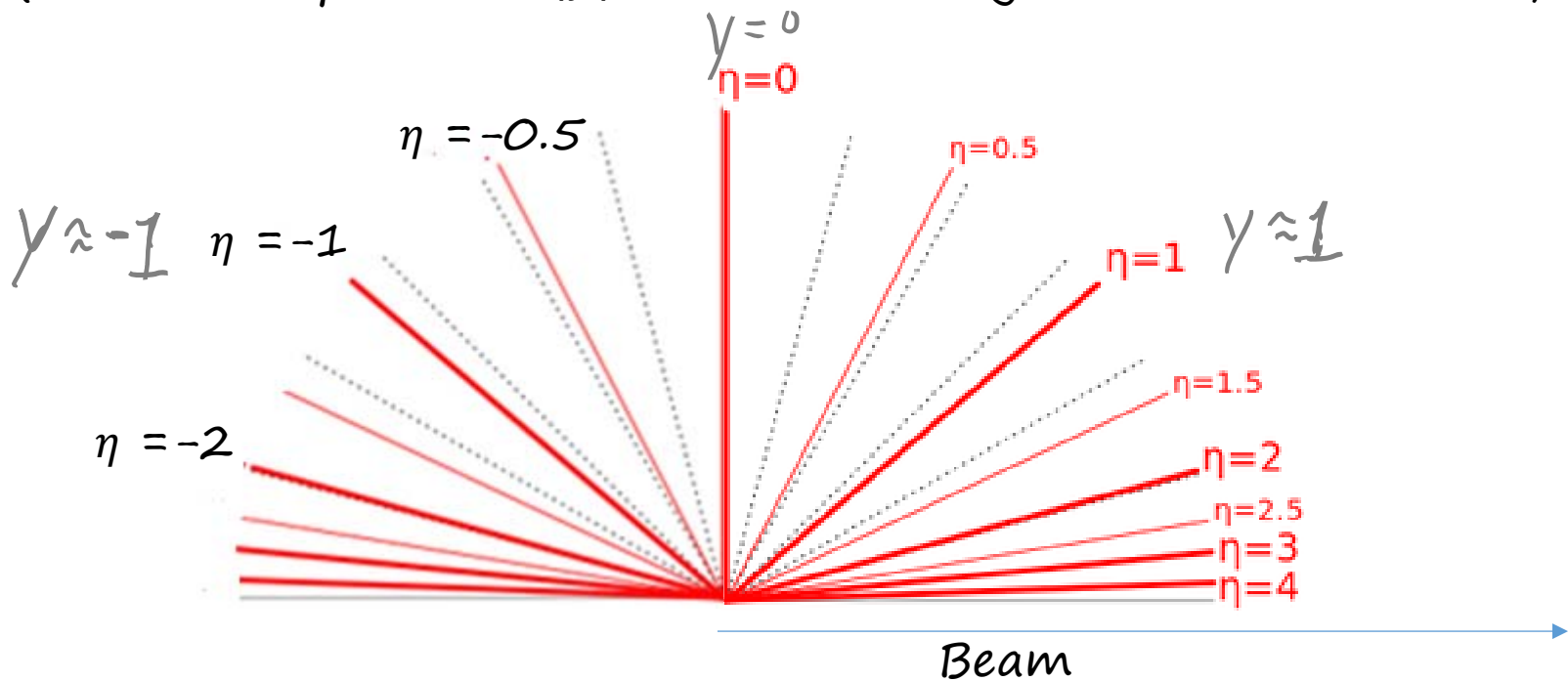

where z is the “global” coordinate z along the beam direction
 Compared to η the magnitude of 3-mom p is replaced by the Energy E

- **Example :** what will y be for a particle moving perpendicular to beam (ie with $\theta = 90^\circ$)
 - In that case $p_z = 0$
 - $y = \frac{1}{2} (\ln (E + 0/E - 0)) = \frac{1}{2} \ln 1 = 0$.
 - also ~same for η



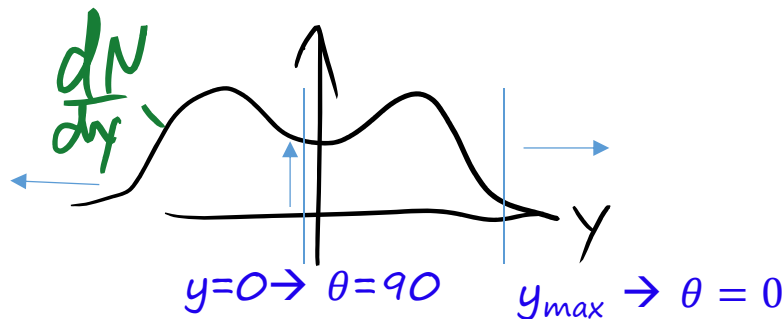
y, η – negative/positive

- So η and y also go from negative to positive symmetrically
- Technically η always can range from $-\infty$ to ∞ ...
- But as we will see, values we actually CARE ABOUT will be limited to $[-n, +n]$ where n is a integer $< \sim 10$...
- [because y (rather than η) will typically be limited to that !]
 - (for massive particles, $|y|$ does not “diverge” ie never reaches ∞)



Important Q: What is y ? Same as η !

- What should we think of **rapidity** as physically?
 - Best to think of it as same as pseudorapidity ! (more “fundamental” thing?)
- Some describe it as like a velocity (because it transforms like classical velocity for Galilean velocity addition as we’ll see next)
- I suggest thinking of it better as just a **direction** more specifically the “ z ” component of a particles direction: (ie like the θ angle—but better relativistically) – ie like η !
 - This is partially because of the more specific way we define it in particle physics w.r.t the beam axis (more on this soon)
 - and also especially because of it’s similarity to pseudo-rapidity (see next and later slides)



Another form of Rapidity Def

- Consider the following “shorter” definition which starts with our previous definition

$$y = \frac{1}{2} \ln \left(\frac{E + p_{\parallel}}{E - p_{\parallel}} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1 + \beta_{\parallel}}{1 - \beta_{\parallel}} \right)$$

special “ β ” (“ $\beta_{\parallel}$ ”)
--equals actual
boost β when p is
completely in “ $\parallel$ ”
direction (for us
the z -direction)

- In both of these versions, it might seem like we could define the direction of the z axis in any direction, but in fact we ONLY consider “parallel” and “transverse” with respect to the BEAM direction. For us, therefore the $p_{\parallel}$ is explicitly the same as p_z , i.e. $p_{\parallel} \equiv p_z$.

Actual Rapidity y Definitions:

- This slide contains several equivalent formulae to define y
- We define y very similarly to η but with E rather than $|p|$

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right)$$

$$y = \frac{1}{2} \ln \left(\frac{1 + p_z/E}{1 - p_z/E} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_z}{1 - \beta_z} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta \cos \theta}{1 - \beta \cos \theta} \right)$$

Because of the above it is also true that for example:

$$y = \operatorname{arctanh}(\beta_z) \quad \text{or} \quad \tanh(y) = \beta \cos \theta$$

this differs from a different but somewhat common definition for rapidity in textbooks of $\tanh y = \beta$ which would only be true for our version for the beam particles themselves (or those with $\theta = 0/180\text{deg}$)

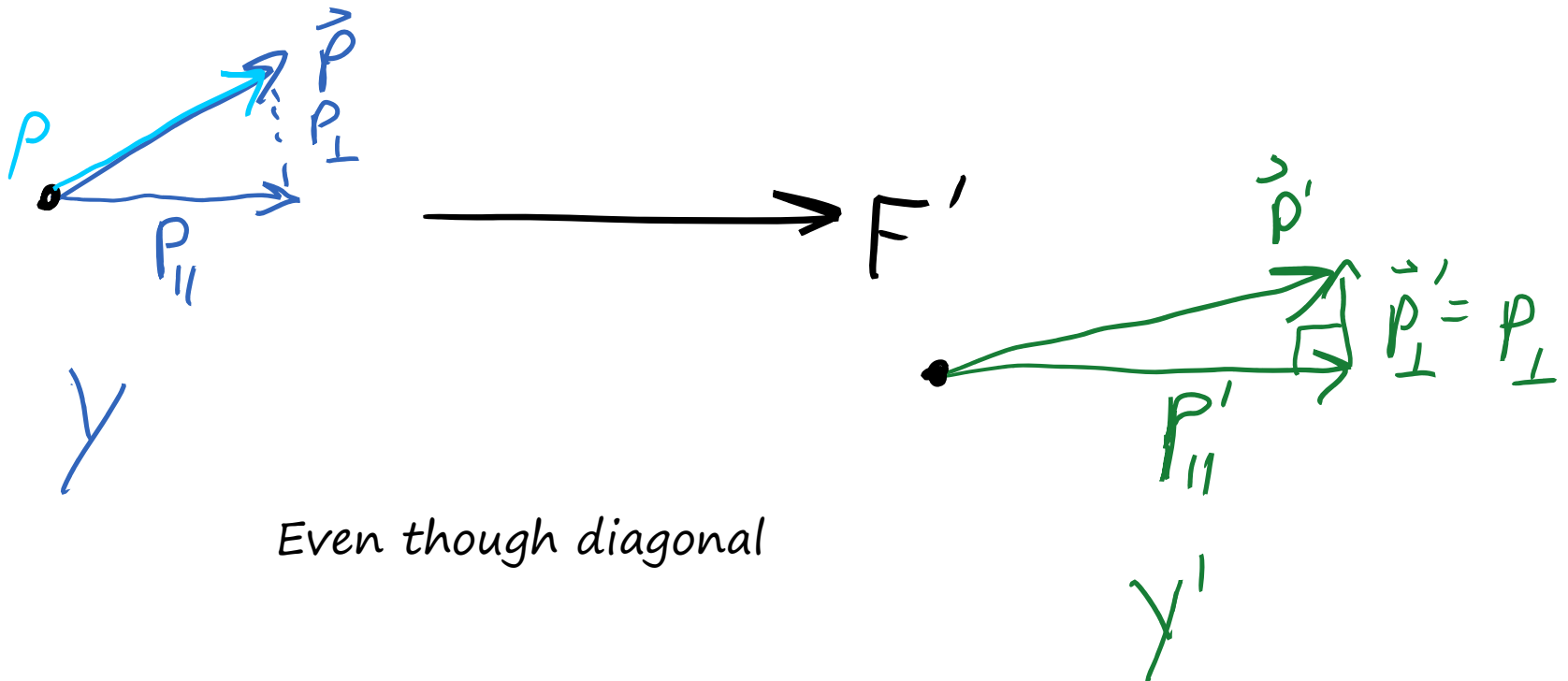
Rapidity

- Consider the following (why? – see next slides)

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

obviously this is either a special " β " (" $\beta_{||}$ ") or is only true when p is completely in " $||$ " direction on z -direction

- Consider y' of a particle in an arbitrary other boosted frame F' (e.g. in frame of another moving particle F' (for us Beam particles in the z -direction) vs rest frame)
- To be clearer what is meant – particle is P



Even though diagonal

• df

Rapidity

- Consider the following (why? – see next slides)

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_{||}}{1 - \beta_{||}} \right)$$

note this is regular β when p is completely in “||” direction on z-direction

- Consider y' of a particle in the boosted frame F' (of the beam particles) vs rest frame using

$$p_{||}' = \gamma (p_{||} + \beta E)$$

$$E' = \gamma (E + \beta p_{||})$$

$$y' = \frac{1}{2} \ln \left(\frac{E' + p_{||}'}{E' - p_{||}'} \right) = \frac{1}{2} \ln \left(\frac{\cancel{\gamma} E + p + \beta(E + p)}{\cancel{\gamma} (E - p) - \beta(E - p)} \right)$$

$$= \frac{1}{2} \ln \left[\left(\frac{E + p_{||}}{E - p_{||}} \right) \left(\frac{1 + \beta_F}{1 - \beta_F} \right) \right] = y + y_{\beta_F}$$

$\frac{1}{2} \ln(\downarrow) + \frac{1}{2} \ln(\downarrow)$

β is boost of F' itself ($\neq p/E$ of particle P , rather $= p_{||F'} / E_{F'}$)
 $y_{\beta} \rightarrow$ rap of $F \equiv y_{F'}$
 & notice therefore it's satisfying the above caveat in red

Again as noted in the colored comments, since we are boosting into the frame F' , the beam particles, which have $\cos\theta = 1$ the actual boost β becomes the β_F that defines the y_{β_F} . We call y_{β_F} the **BEAM RAPIDITY**

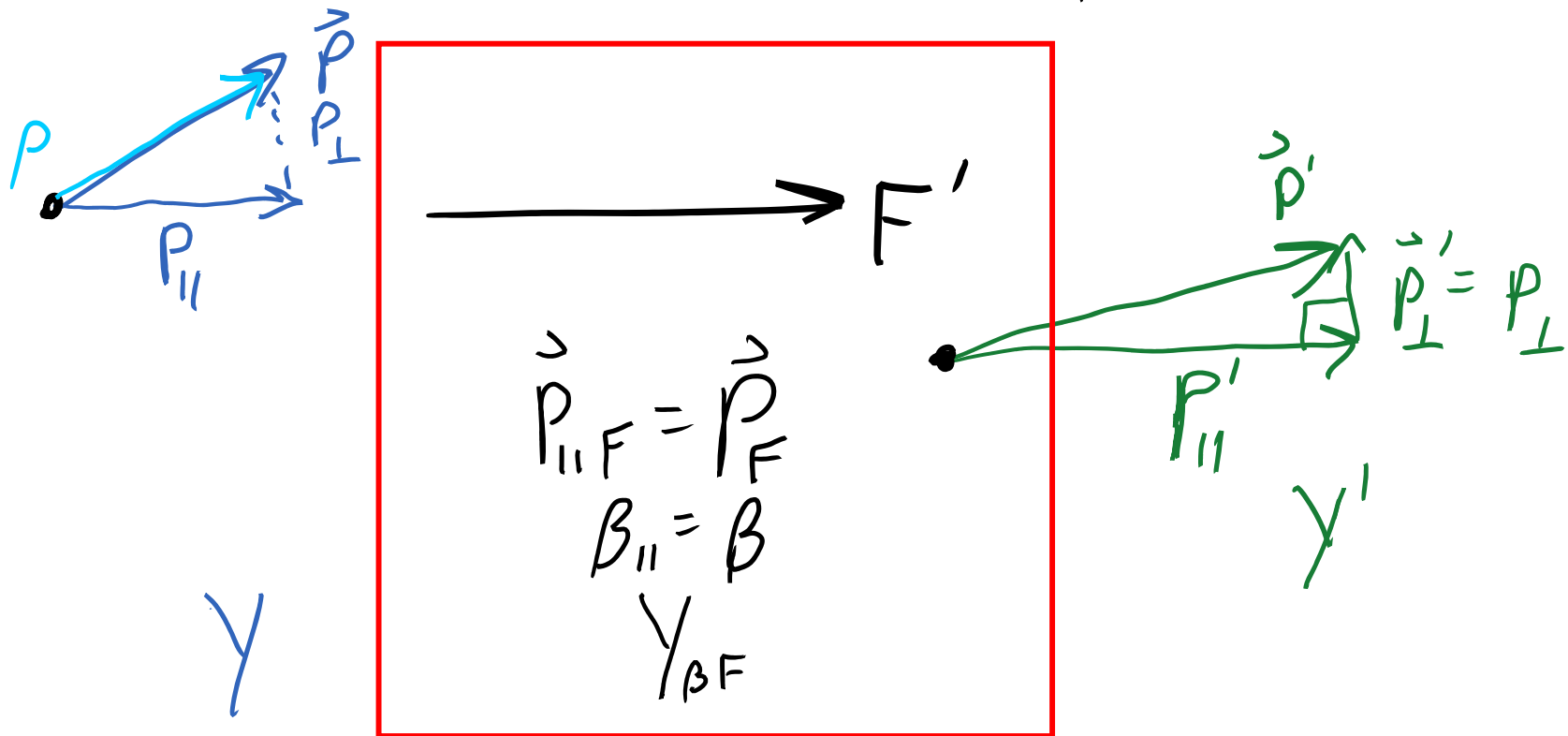
Beam Rapidity Visualization

- Consider the following

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_{||}}{1 - \beta_{||}} \right)$$

$$y' = y + y_{\text{Beam}} \rightarrow y_{\text{BF}}$$

- Visually :



- df

Boosted Rapidity - Implication

- Back to our derivation

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

- Consider y' of a particle in an arbitrary other boosted frame F' (e.g. in frame of another moving particle F') vs rest frame using

$$\begin{aligned} p_{||}' &= \gamma (p_{||} + \beta E) \\ E' &= \gamma (E + \beta p_{||}) \end{aligned} \Rightarrow y' = \frac{1}{2} \ln \left(\frac{E' + p_{||}'}{E' - p_{||}'} \right) = \frac{1}{2} \ln \left(\frac{\cancel{\gamma} (E + p) + \beta (E + p)}{\cancel{\gamma} (E - p) - \beta (E - p)} \right)$$

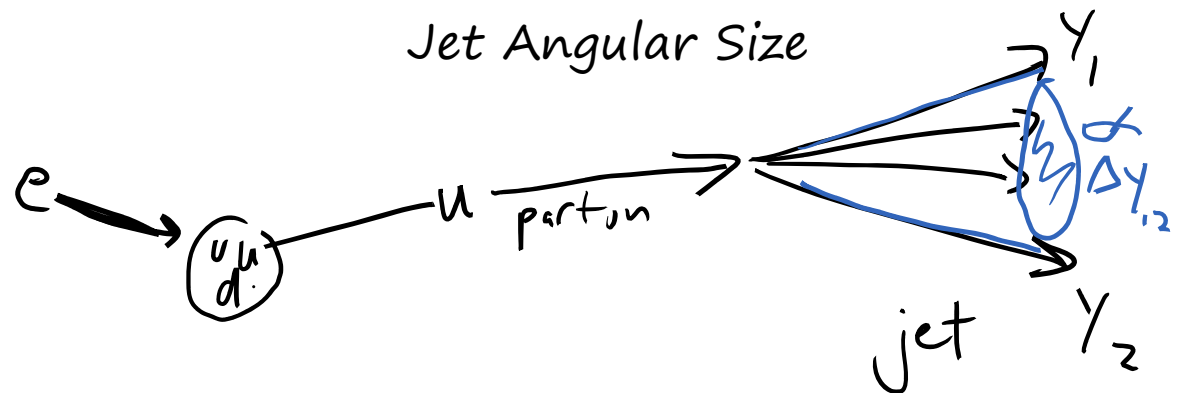
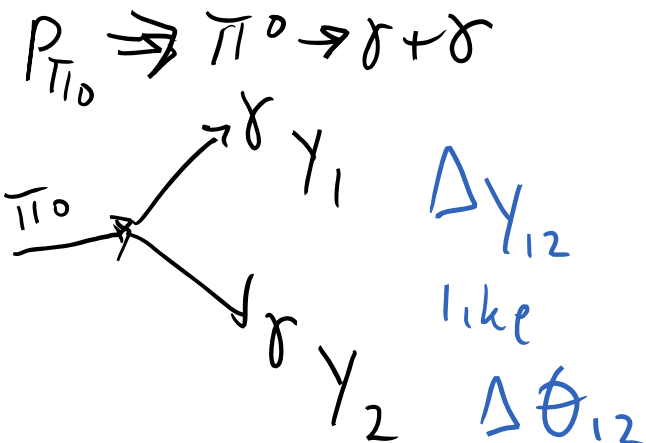
$$= \frac{1}{2} \ln \left[\left(\frac{E + p_{||}}{E - p_{||}} \right) \left(\frac{1 + \beta}{1 - \beta} \right) \right] = \boxed{y + y_{\beta_F}}$$

- Thus boosting rapidity just adds a constant equal to y_{β_F}
 - (dependent on the beam only, same for any particle being transformed into that frame)
- Therefore differences in rapidity in one frame (e.g. between two particles) $\Delta y = y_2 - y_1$ are invariant
- This leads us to first reason why it is useful...

Invariance of Δy

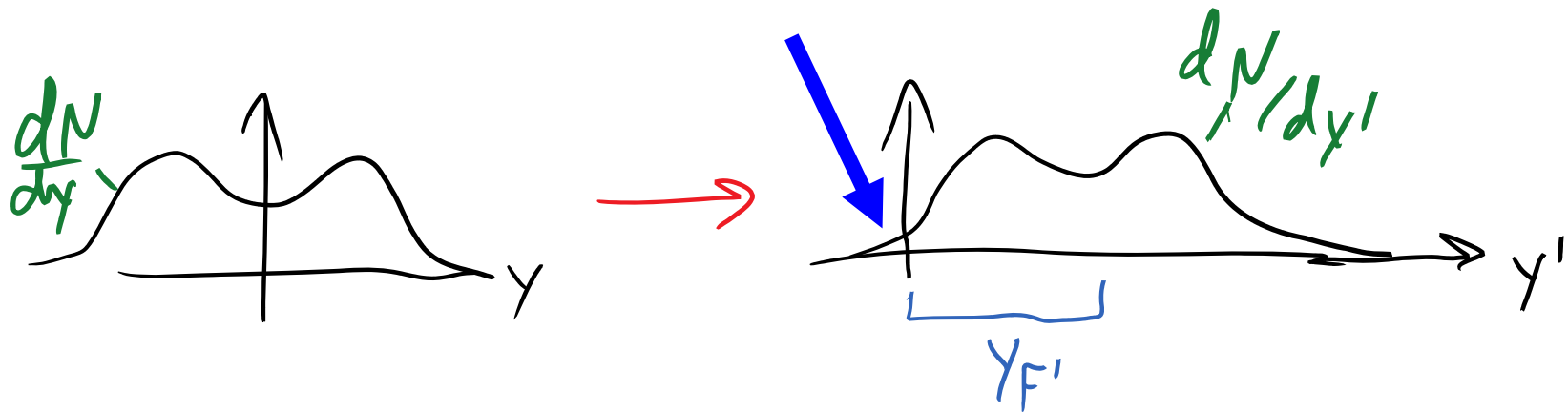
- The significance of $\Delta y = y_1 - y_2$ being invariant means that any (usually angular) “size” scale we define, if expressed in this variable will stay invariant in any frame.
- Furthermore we will show that the numerical size of y and η can be interpreted as the angular size *in radians*.
 - i.e. a Δy of 1, the difference $y_2 - y_1$ of two particles rapidities equaling 1, can be interpreted as them having an angular difference of 1 in their unboosted frame (corresponding to the COM frame/lab frame for colliders) for most cases
- Examples:

π^0 Decay



dy also Invariant

- Amongst many important properties of rapidity the invariance of Δy 's is probably the most important
- Taking the limit $\Delta y = 0 \rightarrow dy$ is also invariant
- This means that (whole) distributions as a function of rapidity ie dN/dy will have the **exact same shape** in boosted frames, just shifted in y by a constant value shift (y_F)



- Ideal e.g. for comparing fixed target vs collider results e.g.
- Recall that y_F in that case would be the beam particle rapidity
- **PRACTICALLY the BEAM PARTICLE RAPIDITY ALSO DEFINES THE LIMIT OF INTERESTING y 's (and therefore η 's) that are interesting.**

Beam Rapidity

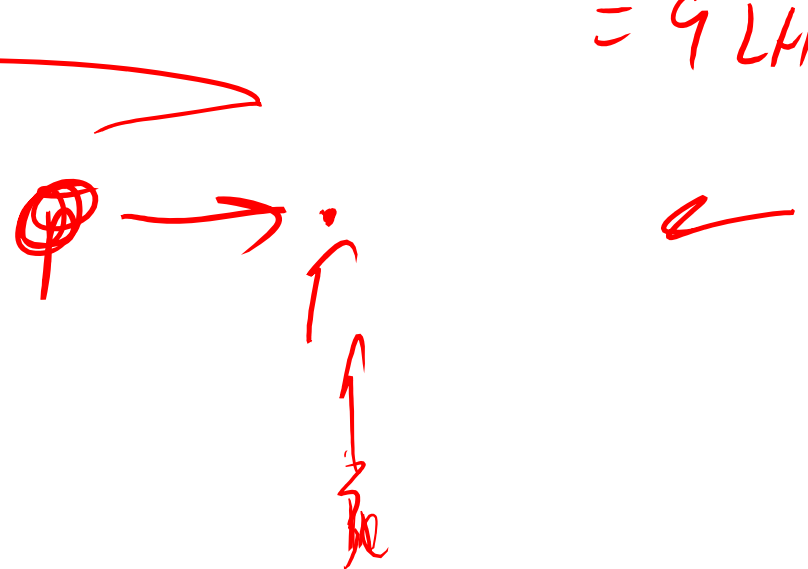
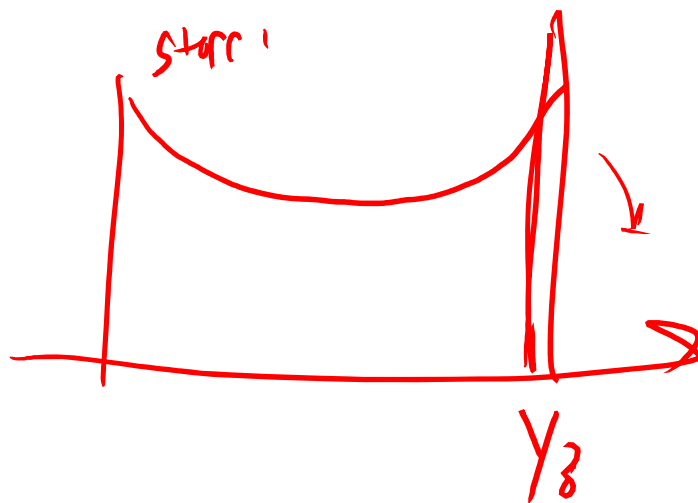
- Beam Rapidity defines the maximum interesting rapidity for most particles:

- $y(E_{\text{beam}}, p_z \text{ beam})$

- E.g. produced hadrons scattered
 - Due to cross sections
 - And Squishyness

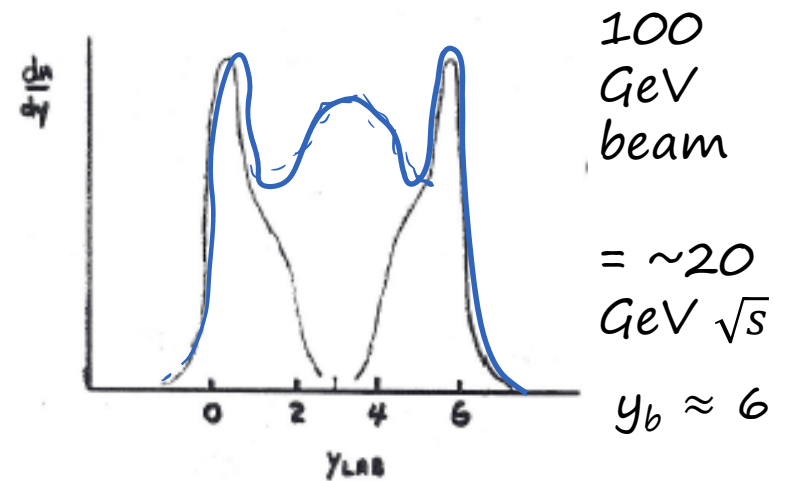
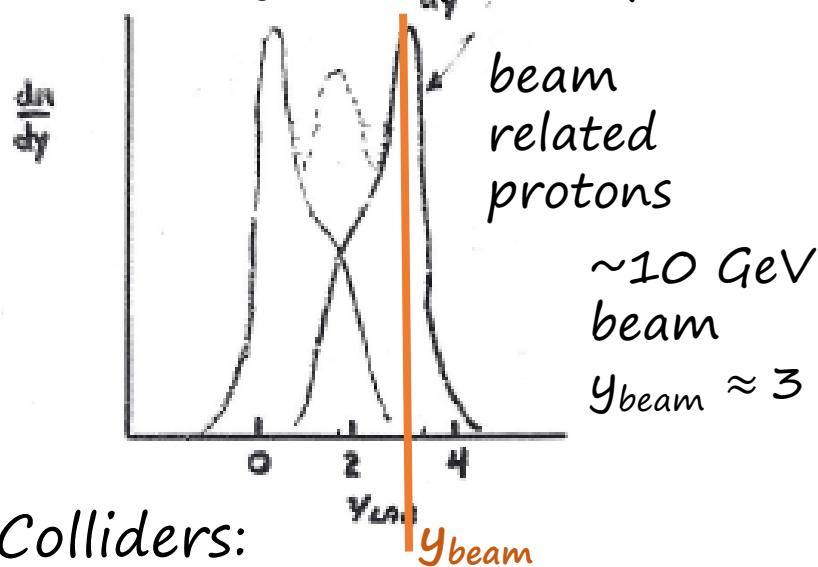
$$y = \ln \left(\frac{E + p_z}{E - p_z} \right) \quad \frac{dn}{dy}$$

y_0
= 6 RHIC
= 9 LHC

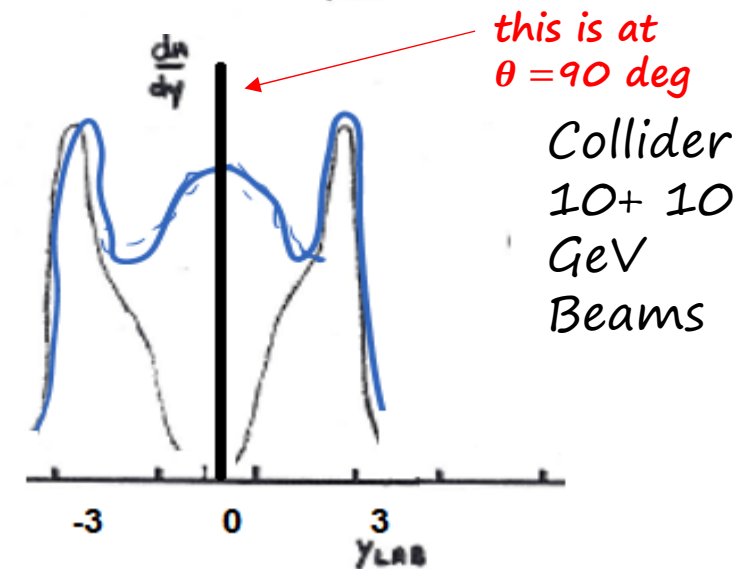
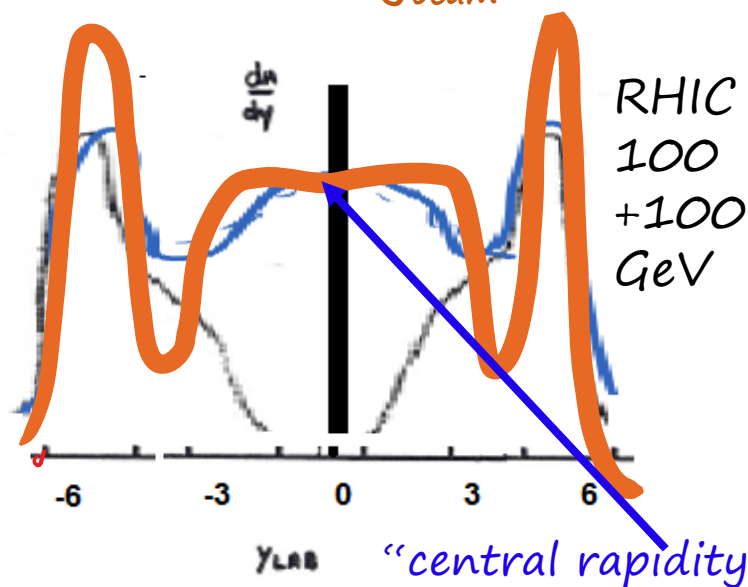


Examples of Particle y Distributions

- Hadron distributions for proton beams (& targets):
- Fixed target (top two plots):



- Colliders:

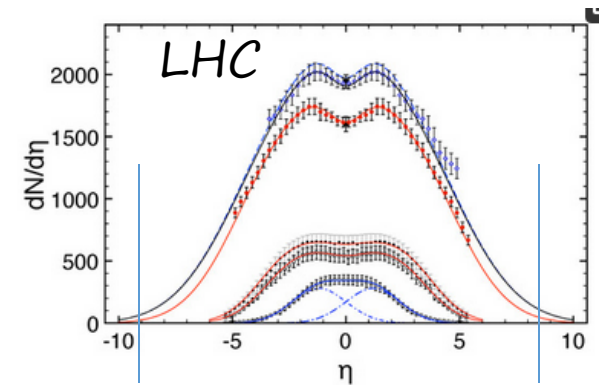
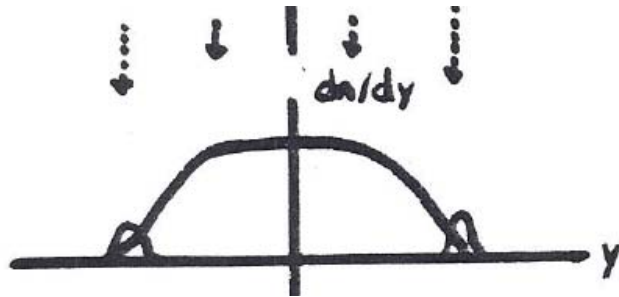


Things to notice about previous slide plots

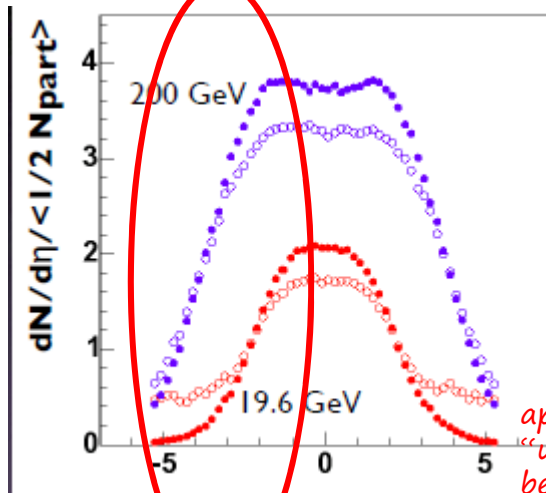
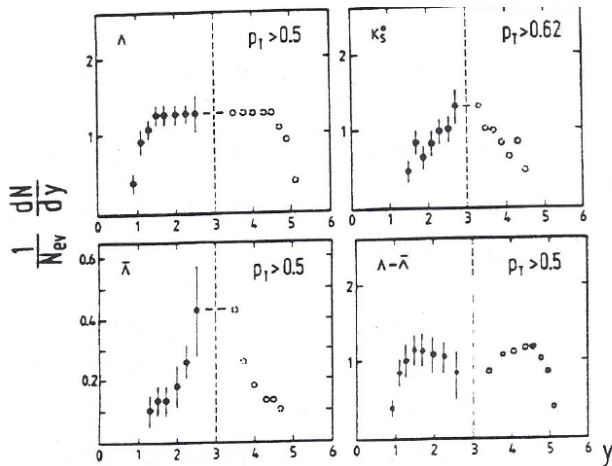
- Each beam energy defines a kind of maximum rapidity value (corresponding to $\theta = \sim 0$ deg wrt beam) and $\sim$ all other particles in event have y values (y -angles) lower than the maximum ($y_{\text{beam}} = 3$: 10 GeV; 6 100 GeV; ($\sim 9-5$ TeV beams at LHC))
 - This makes sense thinking of y as “angle” because angular space get boosted more and more the more p_z the particle has
 - Therefore we can fit more angular ranges in when more energy (and potential higher and higher boosts) available for far forward particles
 - Also as beam energy increases, y -value of dN/dy (dN/dy itself) at center doesn't decrease, distribution mostly just gets wider $\rightarrow$ Extra energy goes into producing more particles to fill up the newly available phase space.
- Comparing **top right** to **bottom right** cases (in green box):, shape of dN/dy is exactly the same, just shifted left. -- in collider case must be symmetric around $y=0 \rightarrow$ this implies fixed target must also be similarly symmetric though not around $y=0$ (instead around $y \approx \frac{1}{2} y_{\text{beam}}$)
 - implying there must be some “activity” around direction “OPPOSITE” to beam in fixed target – note though, this “leftside” peak in that frame does not occur 180 deg from beam direction in fixed target frame.
 - both cases have same total Δy range because they have the same $\sqrt{s}$
 - higher beam energy/beam rap = 6 for one beam in fixed target case (100 GeV) than both beams in collider case (10 GeV+10, $y=3, -3$ —both 3s same as top left fixed target single 10 GeV beam
- Bottom left collider 100 + 100 (look at this last). Double the total Δy ($[-6, 6] \rightarrow 12$ units of rapidity, the $|6|$ is same as for single 100 GeV fixed target beam in top right

Some more typical dN/dy 's , $dN/d\eta$'s

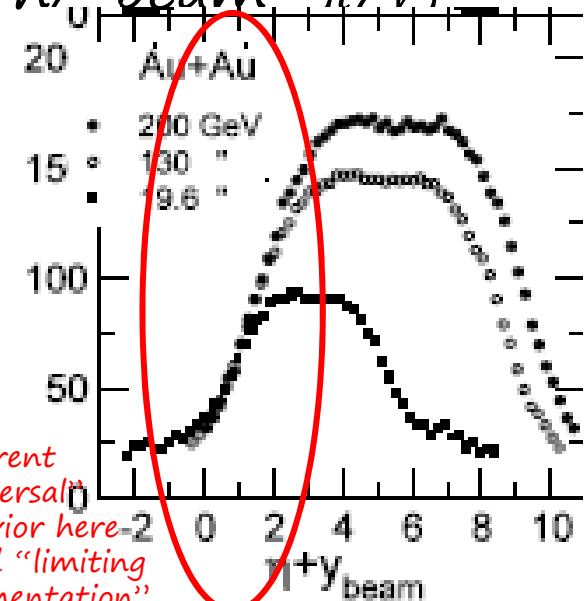
- More realistic even for e.g. protons



- For most particles (not produced pseudoelastically by beam or beam particles themselves)~ (ie not $p/n/$ "beam" π/v)



apparent
"universal"
behavior here -
called "limiting
fragmentation"



Back to Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

In experimental particle physics, **pseudorapidity**, a coordinate describing the angle of a particle relative to the beam axis as

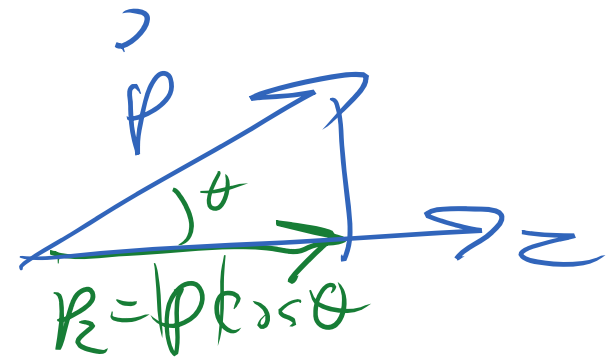
$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the angle between the particle three-momentum and the direction of the beam axis.^[1] Inversely,

$$\theta = 2 \arctan(e^{-\eta}).$$

As a function of three-momentum $\mathbf{p}$, pseudorapidity

$$\eta = \frac{1}{2} \ln \left(\frac{|\mathbf{p}| + p_L}{|\mathbf{p}| - p_L} \right) = \operatorname{artanh} \left(\frac{p_L}{|\mathbf{p}|} \right),$$



$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

- Because of its more direct relation to θ ie GEOMETRY of the detector usually (independent of any particle/momenta) it is probably used in MORE situations

$$|d\eta| = |d\theta|$$

- From the definition of η it is easy to show that at 90 degrees (at $\eta = y = 0$) $d\eta = d\theta$:

$$\eta = -\frac{1}{2} \ln \left(\tan \frac{\theta}{2} \right)$$

$$\frac{d\eta}{d\theta} = -\frac{1}{2} \frac{1}{\tan \theta / 2} \left(\frac{1}{\cos^2 \theta / 2} \right)$$

$\downarrow \quad \theta = 90^\circ \quad \downarrow = 1/1/2$
 $(\frac{\theta}{2} = 45^\circ)$

$$d\eta = -d\theta$$

- This means the scale of $d\eta$ or $\Delta\eta$ is in radians (there @ 90)
- By the boost invariance of $d\eta = dy$, this means its scale is equivalent to radians at ALL y !!!!